

Commercial Trip Cost Estimation 2000-2009

NEFSC Social Sciences Branch Reference Document
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II. Introduction

This document serves as a reference for interpreting the trip cost modeling techniques and cost predictions estimated for northeast commercial fishing trips occurring from 2000-2009. This document gives a brief overview of the econometric modeling techniques and why they are necessary in estimating commercial fishing trip costs as well as the various models generated, their corresponding variables, data cleaning processes, and the timing of annual model re-estimation.

III. Background

Commercial fishing variable costs, (i.e., trip costs) have been collected on a subset of commercial fishing trips in the northeast region for multiple decades. These trip cost data are observed by onboard data collectors who are allocated to vessels to primarily estimate fishing bycatch rates. Over time, the method in which observers have been allocated has changed. From 2010 to present¹ day, observer sea day allocation has been stratified using the Stratified Bycatch Reporting Methodology (SBRM). From 2005-2009 NEFOP observers are stratified using a slightly less sophisticated method which eventually evolved into the SBRM strata known today. Prior to 2005, the method for allocating observers was ad hoc and based on Northeast Fisheries Science Center staff discussions. Ultimately, observer coverage has always been stratified on biological data needs rather than economic which can create bias in cost estimates. Bias in trip cost estimates was explored in a publication authored by SSB and WHOI staff in a 2020 publication² where Heckman Selection models were used to simultaneously test and correct for selection bias in trip cost estimation. Results of the study find that the majority of trip cost models investigated suffer from selection bias. As described by the manuscript, Ordinary Least Squares (OLS) and Heckman mean model predictions differ by about 8%, on average, and range from 1-42% across models, suggesting modeling choice could result in moderately different conclusions at the aggregate level but could yield drastically different results when analyzing costs at the sub-fleet level. Furthermore, discrepancies across the model distributions tend to be greatest for larger cost predictions which can lead to errors in assessing or reporting trip cost maxima. Here, we extend the findings of the manuscript in an effort to generate

¹ In 2009, a mix of SBRM and the 2005 stratification methodology was used to allocate observer sea-days which is why it is not included in this series.

² Werner, S., DePiper, G., Jin, D. and Kitts, A., 2020. Estimation of Commercial Fishing Trip Costs Using Sea Sampling Data. *Marine Resource Economics*, 35(4), pp.379-410.

unbiased trip cost models and predict costs for all northeast commercial fishing trips completed during 2000-2009.³

IV. Database Generation

The project database was compiled using information from two main sources: (i) vessel trip reports (VTR) and (ii) observer data containing realized trip cost information. The VTR records were used to identify all commercial fishing trips accomplished during 2000-2009. Observer data, containing trip cost information, were merged into the VTR dataset, creating a database of all trips taken during this time period—some of which carried an onboard data collector who recorded trip cost data while the remainder of the trips in the dataset have no trip cost record.⁴ Average weekly wages of marina workers and New York Harbor diesel prices were taken from secondary sources and were used in the trip cost models.⁵ Vessel characteristics (e.g., age, horsepower and gross tons) were retrieved from the Permit database. All other variables in the analysis were provided via the VTR data records and were retained in the full project dataset. Primary gear group (Appendix I) was identified for each trip which landed the highest quantity of catch (lbs.).

V. Model Types: Gears and Trip Durations

According to the trip's primary gear, trips were grouped into general gear categories where gear groups are believed to have similar cost functions. The gear groups for modeling include: trawl, gillnet, dredge, and a multiple gear group. The multiple gear types group includes longline, seine, pots and traps, handlines and handgear/other gear.⁶ The multiple gear group category was pooled given that handgear and "other" gear only had 18 observed trips over the 9 year time period. A Chow Test was implemented to identify if a structural break exists between seine, longline, and pot and trap models where results were

³ Prior to 2010, different methods were used to stratify onboard observers such that controlling for bias generated via SBRM stratification, such that slightly different methods were used for generating cost estimates for trips taken during 2000-2009. Information on the 2005-2009 stratification can be found here Rago PJ, Wigley SE, Fogarty MJ. 2005. NEFSC bycatch estimation methodology: allocation, precision, and accuracy. Northeast Fish Sci Cent Ref Doc. 05-09; 44 p. Online at: <https://repository.library.noaa.gov/view/noaa/3455>

⁴ The observer records were merged into the VTR database using a hierarchical matching algorithm based on the vessel hull number, gear type, fishing area, and date sailed/landed. The VTR serial number, which is assigned i) at the beginning of a fishing trip ii) when a trip changes statistical area or iii) when a vessel changes the gear on a fishing trip, was used as a separate merging variable following the hierarchical merge to ensure all possible trips which carried an onboard data collector were matched with their correct VTR record.

⁵ Marina worker wage data was retrieved from the Quarterly Census of Employment and Wages page of the Bureau of Labor Statistics website: <https://www.bls.gov/cew/datatoc.htm>. Diesel price information from the Department of Energy website: https://www.eia.gov/dnav/pet/hist/er_epr2dxl0_pf4_y35ny_dpgD.htm.

⁶ "Other" gear includes weirs, diving gear, harpoons, rakes, Fyke nets, and a large proportion of trips using NEGEAR 999 (Unknown gear).

not significant across all variables, (not including the constant), failing to reject the null hypothesis. This grouping allowed for the pooling of these gear types into a single model.

i. Selection Bias and Trip Cost Predictions

The Heckman Selection model depends on 1.) an underlying linear regression and 2.) a selection equation comprised of variables driving selection bias. In cases where the Heckman Selection model fails to reject the null hypothesis ($\rho = 0$), the selection equation becomes unnecessary and an OLS model was used to estimate costs, as OLS is still BLUE (*See section V. subsection iv. Selection Bias and Trip Cost Predictions*). The dependent variable in models (either the Heckman or OLS) is the natural log of a composite trip cost variable comprised of the cost of fuel, ice, bait, food/groceries, water, oil and supplies and is presented in 2009 constant dollars (*see section V. subsection ii. Dollar Values, below*). Table 1 describes the independent variables used in the 2000-2009 trip cost models.

The selection equation for this time series differs from that used to estimate models post 2010, given that they were stratified differently between these time periods. The selection equation for these models includes the number of employed observers by month as a proxy for observer effort as well as a gear variable. In the trawl, gillnet, and dredge equations, an indicator variable for NEGEAR⁷ was used in the selection equation given that the NEGEAR types often link to targeted species which is expected to be linked to the past ad hoc stratification methods as well as the modified SBRM methods used over 2000-2009. NEGEAR was not added to the main regression for these gear groups given that the distinction is not expected to impact variable costs, simply the types of trips selected for observation. In the multiple gear model, a gear group indicator variable was included in both the selection equation as well as the main regression to control for both the impact on costs as well as the probability of selection. The NEGEAR variable was not used in the multiple gear model due to the low level of variation of observed NEGEAR types.

Each gear-type/trip duration was initially modeled using the Heckman Selection method to test for selection bias. If the Heckman Selection model results suggest the presence of selection bias the Heckman model was used to predict trip costs. If the Heckman selection model failed to reject the null hypothesis ($\rho = 0$) then selection bias is not significant and an OLS model was used for trip cost prediction. The Wald Test of Independent Equations as well as model choice for generating 2000-2009 trip cost predictions are presented in Table 2.

ii. Model Dollar Values

⁷ See Appendix I for NEGEAR definitions.

All variables were adjusted to 2009 constant US dollars using the GDP implicit price deflator (base = average by quarter). This includes the composite trip cost variable, the price of diesel, and the average weekly wage of marina workers (<https://fred.stlouisfed.org/series/GDPDEF>). This is to control of inflation directly within the model. Cost predictions are also presented in nominal dollars for convenience.

iii. Clustered Standard Errors

Heteroscedasticity was identified in the MRE manuscript models using the Breusch-Pagan/Cook-Weisberg Test for Heteroskedasticity. Standard errors were clustered on either the seasonal quarter (QTR) or the primary fishing region⁸ to address Heteroscedasticity in all Heckman and OLS models.

iv. Retransformation Process

As the independent variable (trip cost) was log transformed for modeling purposes, Duan's smearing estimate (1983) was used as a retransformation process post trip cost prediction.⁹

⁸ The fishing region is a categorical variable consisting of the Gulf of Maine, Georges Bank or "other" based on the trip's CAREA which had the highest lbs. of kept catch for each trip.

⁹ Duan, N. 1983. "Smearing estimate: a nonparametric retransformation method." *Journal of the American Statistical Association* 78:605-610.

Table 1. Ordinary Least Squares (OLS) and Heckman Selection Independent Variable Descriptions

Independent Variable	Description	Units	Model*
Ln Vessel Horse Power	Natural log of vessel engine horsepower.	kW	O/H, HS
Ln Vessel Gross Tons	Natural log of vessel weight in gross metric tons.	Tons	O/H, HS
Ln Vessel Age	Natural log of the age of vessel (from when the vessel was built to calendar year 2009).	Years	O/H, HS
Ln Hours Absent	Natural log of the hours the vessel was absent from port (from home port to destination) during a commercial fishing trip.	Hours	O/H, HS
Ln Diesel Price	Natural log of the price of Ultra-Low Sulfur Number 2 diesel (New York Harbor) adjusted to 2009 constant U.S. dollars. Obtained from Department of Energy website.	\$/Gallon	O/H, HS
Ln Average Weekly Wage	Natural log of the average weekly wages for marina workers by state as an approximation of crew member opportunity costs, adjusted to 2009 constant U.S. dollars. Obtained from the Quarterly Census of Employment and Wages page of the Bureau of Labor Statistics website.	\$/Week	O/H, HS
Seasonal Quarter	Categorical variable specifying the season in which the fishing trip commenced using three-month time periods within a financial calendar (Q1, Q2, Q3, Q4).	Categorical Variable	O/H, HS
Calendar Year	Calendar year in which the fishing trip was completed.	Categorical Variable	O/H, HS
NEGEAR^	Gear variable which describes the gear types along with the species targeted.	Categorical Variable	HS
Gear Group^	Gear group variable that categorizes the primary gear used on the trip based on quantity of catch landed per trip. Gear groups include: dredge, gillnet handgear, longline, pots and traps, seine, trawl and unknown.	Categorical Variable	O/H, HS

Day/Multi Day Trip	Indicator variable of a day trip (≥ 24 hours) or multi-day trips (< 24) hours.	Categorical Variable	O/H, HS
Ln Number of Observers	The natural log of the number of onboard vessel observers employed during the month and year that the trip was taken.	Numeric Value	HS

**Describing the models in which the independent variable is incorporated, where O/H = OLS or Heckman Selection model (Trip cost regression) and the HS = Heckman Selection model (Selection Equation).*

^ The NEGEAR indicator variable was only included in the selection equation of trawl, dredge, and gillnet given that they have higher levels of observations across NEGEARs which is expected to impact the selection based on the NEGEAR type within the gear group. The multiple gear model uses the courser gear group definition and is contained in the selection and regression portion of the Heckman model given that the gear types are pooled and the gear types are expected to change the cost equation intercept given the pooled nature of this equation.

Table 2. Model types and estimation techniques used for 2000-2009 commercial fishing trip cost predictions

Gear Type	No. Observed Trips ^a	No. Unobserved Trips ^a	Wald Test of Independent Equations		Model Used for Trip Cost Prediction
			χ^2	$P > \chi^2$	
Trawl	9,066	354,466	1.75	0.1862	OLS
Dredge	2,705	136,146	121.63	0.000***	Heckman Selection
Gillnet	6,737	173,071	440.65	0.000***	Heckman Selection
Multiple Gears	949	568,518	N/A	N/A	OLS*

** The small number of observed trips generated high condition numbers in the Heckman Selection model which made running the model unfeasible. OLS was implemented in this case.*

**** Indicates significance at the 1% level.*

^a Number of observed and unobserved trips used in the cost model.

VI. Data Cleaning

In creating the composite trip cost variable, the average annual fuel price was used for any trip that reported a fuel price less than or equal to zero (~10% of Observer Database Records). Additionally, any record reporting fuel gallons less than or equal to zero was replaced with a predicted gallon value based on hourly trip duration. The mean annual ice price was used for any trip missing the price of ice but reported a positive value for ice tons used. All other missing values were interpreted as a zero cost. Of all VTR trip records, there were a few cases where the hours reported were less than or equal to zero. In these cases, the median hourly trip duration value was used for trip cost prediction.

VII. Additional Variables Considered

A regional variable was considered, but was not added as a selection variable given that it is unclear how NEFSC staff members selected vessels for observation outside of the gear types prior to 2005. For the 2010-present day models, departure state indicator variables were considered to control for state-level price variations in the cost estimation equation. These variables were tested across each of the models using OLS using data from 2010-2021. OLS results suggested there were few states which were significant within each model, if any, and added little to the overall explanatory power of the models. These parameters also caused the condition number within the Heckman selection model to increase, many times above the acceptable threshold. It was for this reason that state variables were not included in the regressions.

VIII. Outliers

Outliers were addressed during various stages of the modeling and cost prediction process. Each of the seven trip cost sub-categories (fuel, ice, water, bait, oil, groceries and supplies) were Winsorized individually prior to generating the composite trip cost variable.¹⁰ The composite trip cost value was also Winsorized by gear and trip duration prior to modeling such that cost model estimates were not impacted by outliers. Trips which reported durations less than 12 minutes or more than 20 hours were removed from the modeling stage. Lastly, trip cost model predictions are presented in two forms, both Winsorized and not Winsorized, where the raw predictions are the values estimated using the models described above which yield some unreasonably high or low cost predictions—often driven by unlikely/erroneous trip durations (hours) reported in the VTR database. This is an ongoing QA/QC issue which is not easily resolvable. The Winsorized trip costs are presented as a method of avoiding unreasonably high or low trip costs, replacing any value within each gear-group that is less than the 1st percentile or greater than the 99th percentile with the 1st and 99th percentile value, respectively (winsor2 in stata). The raw values are presented for the end-user's convenience, such that a different method of outlier correction can be used if desired; however, extreme values should be noted when using the unWinsorized trip cost predictions.

IX. Trip Cost Values and Predictions (2000-2009) and data usage

Trip cost values and predictions for 2000-2009 are presented in the Social Sciences network file, pathway: \\net.nefsc.noaa.gov\socialsci\Trip_Costs\2000-2009\ 2010_2009_Commercial_Fishing_Trip_Costs. Trip cost predictions were generated using the modeling methods indicated in Table 2. If the actual

¹⁰ Winsorizing replaces any value less than the 1st or greater than the 99th percentile with the 1st and 99th percentile value, respectively.

cost of a trip was recorded for a specific trip (i.e., collected by an onboard data collector) that cost is presented in the 2000-2009 Trip Costs Excel file rather than a prediction. Trip costs in the 2000-2009 Trip Costs Excel file are presented in 2009 U.S. Dollars (*see Table 1 on variable descriptions and V. subsection ii. Dollar Values*) as well as nominal values. Costs estimates were retransformed using Duan's smearing estimator (1983) (*see section V. subsection v. Retransformation Process*). The VTR Trip ID (TRIPID) and landing year¹¹ is presented in the Excel file along with the trip cost predictions which allows trips to be linked back to their original VTR record. Note, TRIPIDS should be unique across VTR trip databases however, this is not always the case (due to QA/QC errors) such that the landing year variable is necessary to create a unique link back to the VTR source. When using the data, it is requested that the end-user notify Samantha Werner (Samantha.Werner@noaa.gov) for reporting and tracking purposes. If there are suggestions to improving these models or general questions, please direct them to Samantha Werner.

Table 3. Trip cost Excel file variables and descriptions

Variable Name	Variable Description
VTR_TRIPID	Vessel trip report trip id
DB_LANDING_YEAR	The year in which the trip landed according to the VTR database. Specifically, the VTR DATELAND1 variable from the VTR.VESLOGxxxxt data tables.
TRIP_COST_2009_DOL	Estimated and reported trip costs in 2009 constant U.S. dollars
TRIP_COST_WINSOR_2009_DOL	Estimated and reported trip costs in 2009 constant U.S. dollars, winsorized at the 1st and 99th percentiles where costs greater than or less than these percentiles were replace with the 1st and 99th percentile values.
OBSERVED_COST_DUMMY	Indicator variable of whether or not the trip had their trip costs recorded by an onboard observer (0 = unobserved, 1= observed)
TRIP_COST_NOMINALDOLS	Estimated and reported trip costs in nominal U.S. dollars
TRIP_COST_NOMINALDOLS_WINSOR	Estimated and reported trip costs in nominal U.S. dollars, winsorized at the 1st and 99th percentiles where costs greater than or less than these percentiles were replace with the 1st and 99th percentile values.

¹¹ The year in which the trip landed according to the VTR database. Specifically, the VTR DATELAND1 variable from the VTR.VESLOGxxxxt data tables.

APPENDIX I . NEGEAR and General Gear Group Descriptions

NEGEAR	Gear Name	Gear Group
10	LOGLINE, BOTTOM	Longline
20	HANDLINE	Handgear
21	HANDLINE,AUTO JIG	Handgear
30	HARPOON,OTHER/NK SPECIES	Handgear
31	HARPOON,SWORDFISH	Handgear
32	HARPOON,TURTLE	Handgear
34	SPEARS	Handgear

40	LOGLINE, PELAGIC	Longline
50	TRAWL,OTTER,BOTTOM,FISH	Trawl_Bottom
51	TRAWL,OTTER,BOTTOM,CRAB	Trawl_Bottom
52	TRAWL,OTTER,BOTTOM,SCALLOP	Trawl_Bottom
53	TRAWL,OTTER,BOTTOM,TWIN	Trawl_Bottom
54	TRAWL,OTTER,BOTTOM,RUHLE	Trawl_Bottom
55	TRAWL,OTTER,BOTTOM,LOBSTER	Trawl_Bottom
56	TRAWL,OTTER,BOTTOM,PAIRED	Trawl_Bottom
57	TRAWL,OTTER,BOTTOM,HADDOCK SEPARATOR	Trawl_Bottom
58	TRAWL,OTTER,BOTTOM,SHRIMP	Trawl_Bottom
59	TRAWL,OTTER,BOTTOM,OTHER/NK SPECIES	Trawl_Bottom
60	TROLL LINE, OTHER/NK SPECIES	Handgear
61	TROLL LINE, TUNA	Handgear
62	TROT LINE WITH BAIT	Handgear
63	POWER TROLL SALMON	Handgear
64	POWER TROLL TUNA	Handgear
65	POWER TROLL OTHER/NK SPECIES	Handgear
66	TROLL & HANDLINE, COMBINED	Handgear
67	TROLL LINE, GREEN-STICK	Handgear
70	HAUL SEINE, BEACH, COMMON	Seine
71	HAUL SEINE, LONG	Seine
80	FLOATING TRAP	Pot/Trap
90	DIP NET, COMMON	Net
91	DIP NET, DROP NET	Net
92	DIP NET, PUSH NET	Net
	GILL NET, FIXED OR ANCHORED,SINK, OTHER/NK	
100	SPECIES	Gillnet
101	GILL NET,SET /STAKE, SEA BASS	Gillnet
102	GILL NET,STAKE,OTHER	Gillnet
103	GILL NET,SET, SALMON	Gillnet
104	GILL NET,SET, CRAB	Gillnet
105	GILL NET,ANCHORED-FLOATING, FISH	Gillnet
110	GILL NET,DRIFT, OTHER/NK SPECIES	Gillnet
111	GILL NET,DRIFT, SEA BASS	Gillnet
112	GILL NET,DRIFT, SHAD	Gillnet
113	GILL NET,DRIFT, SALMON	Gillnet
114	GILL NET,DRIFT, BARRACUDA	Gillnet
115	GILL NET,DRIFT, LARGE PELAGIC	Gillnet
116	GILL NET,DRIFT-FLOATING, FISH	Gillnet
117	GILL NET,DRIFT-SINK, FISH	Gillnet
120	PURSE SEINE, OTHER/NK SPECIES	Seine
121	PURSE SEINE, HERRING	Seine
122	PURSE SEINE, MACKEREL	Seine
123	PURSE SEINE, MENHADEN	Seine
124	PURSE SEINE, TUNA	Seine
131	DREDGE, SCALLOP,BAY	Dredge_Scallop
132	DREDGE, SCALLOP,SEA	Dredge_Scallop
140	POUND NET, OTHER/NK SPECIES	Net
141	POUND NET, CRAB	Net
142	POUND NET, FISH	Net
143	POUND NET, HORSESHOE CRAB	Net

150	TRAWL,OTTER,BOTTOM,LG MESH BELLY PANEL	Trawl_Bottom
160	DANISH SEINE	Seine
170	TRAWL,OTTER,MIDWATER PAIRED	Trawl_Midwater
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
181	POTS + TRAPS,FISH	Pot/Trap
182	POTS + TRAPS,EEL	Pot/Trap
183	POTS + TRAPS,CONCH	Pot/Trap
184	POTS + TRAPS,OCTOPUS	Pot/Trap
185	POTS + TRAPS,TURTLE	Pot/Trap
186	POTS + TRAPS, HAGFISH	Pot/Trap
190	POTS + TRAPS,SHRIMP	Pot/Trap
200	POT/TRAP, LOBSTER OFFSH NK	Pot/Trap
201	POT/TRAP, LOBSTER OFFSH WOOD	Pot/Trap
202	POT/TRAP, LOBSTER OFFSH WIRE	Pot/Trap
203	POT/TRAP, LOBSTER OFFSH WD/WR	Pot/Trap
204	POT/TRAP, LOBSTER OFFSH WD/NET	Pot/Trap
205	POT/TRAP, LOBSTER OFFSH WR/NET	Pot/Trap
206	POT/TRAP, LOBSTER OFFSH PLASTIC	Pot/Trap
207	POT/TRAP, LOBSTER OFFSH CONICAL	Pot/Trap
210	POT/TRAP, LOBSTER INSH NK	Pot/Trap
211	POT/TRAP, LOBSTER INSH WOOD	Pot/Trap
212	POT/TRAP, LOBSTER INSH WIRE	Pot/Trap
213	POT/TRAP, LOBSTER INSH WD/WR	Pot/Trap
214	POT/TRAP, LOBSTER INSH WD/NET	Pot/Trap
215	POT/TRAP, LOBSTER INSH WR/NET	Pot/Trap
216	POT/TRAP, LOBSTER INSH PLASTIC	Pot/Trap
217	POT/TRAP, LOBSTER INSH CONICAL	Pot/Trap
220	HOES	Handgear
221	FORKS	Handgear
222	SHOVELS	Handgear
223	PICKS	Handgear
230	BY HAND, OTHER/NK SPECIES	Handgear
231	BY HAND, OYSTERS	Handgear
240	STOP SEINE	Seine
250	RAKES, OTHER/NK SPECIES	Handgear
251	RAKES, OYSTER	Handgear
260	WEIR	Handgear
270	BOX TRAP	Pot/Trap
281	DREDGE, OYSTER COMMON	Dredge_Scallop
282	DREDGE, OYSTER SUCTION	Dredge_Scallop
290	BAG NET	Net
300	POTS + TRAPS,CRAB OTHER	Pot/Trap
301	POTS + TRAPS,BLUE CRAB	Pot/Trap
310	LIFT NET	Net
320	FYKE NET, OTHER/NK SPECIES	Net
321	FYKE NET, CRAB	Net
322	FYKE NET, FISH	Net
323	FYKE NET, TURTLE	Net
324	HOOP NET	Net
330	DIVING GEAR	Handgear
331	DIVING GEAR, SCALLOP	Handgear

332	DIVING GEAR, URCHIN	Handgear
334	DIVING GEAR,SUCTION PUMP,HAND	Handgear
340	CAST NETS	Handgear
350	BEAM TRAWL, OTHER/NK SPECIES	Trawl_Bottom
351	BEAM TRAWL, SHRIMP	Trawl_Bottom
352	BEAM TRAWL,SCALLOP	Trawl_Bottom
353	BEAM TRAWL,FISH	Trawl_Bottom
360	SCOTTISH SEINE	Seine
370	TRAWL,OTTER,MIDWATER	Trawl_Midwater
380	SCRAPES	Handgear
381	DREDGE, OTHER/NK SPECIES	Dredge_Scallop
382	DREDGE, CLAM	Dredge_Scallop
383	DREDGE, CRAB	Dredge_Scallop
384	DREDGE, CONCH	Dredge_Scallop
385	DREDGE, MUSSEL	Dredge_Scallop
386	DREDGE, CLAM, HYDRAULIC	Dredge_Hydraulic
387	DREDGE, URCHIN	Dredge_Scallop
390	STOP NET	Net
400	DREDGE, SURF CLAM + OCEAN QUAHO	Dredge_Hydraulic
410	TONGS AND GRABS, OTHER/NK SPECIES	Handgear
411	TONGS AND GRABS, CLAM	Handgear
412	TONGS AND GRABS, OYSTER	Handgear
413	TONGS AND GRABS, CLAM,PATEN	Handgear
414	TONGS AND GRABS,OYSTER,PATEN	Handgear
420	HOOKS	Handgear
430	KNIFE, SEAWEED	Handgear
440	WEEDWHACKER, SEAWEED	Handgear
450	TRAWL,SHRIMP,TWINNED	Trawl_Bottom
500	GILL NET,RUNAROUND	Gillnet
510	GILL NET,GREAT LAKES	Gillnet
520	GILL NET,OTHER	Gillnet
530	TRAMMEL NET	Net
810	SUCTION PUMP	Handgear
999	UNKNOWN	Unknown
10	LOGLINE, BOTTOM	Longline
20	HANDLINE	Handgear
20	HANDLINE	Handgear
21	HANDLINE,AUTO JIG	Handgear
21	HANDLINE,AUTO JIG	Handgear
34	SPEARS	Handgear
34	SPEARS	Handgear
40	LOGLINE, PELAGIC	Longline
40	LOGLINE, PELAGIC	Longline
50	TRAWL,OTTER,BOTTOM,FISH	Trawl_Bottom
50	TRAWL,OTTER,BOTTOM,FISH	Trawl_Bottom
59	TRAWL,OTTER,BOTTOM,OTHER/NK SPECIES	Trawl_Bottom
59	TRAWL,OTTER,BOTTOM,OTHER/NK SPECIES	Trawl_Bottom
59	TRAWL,OTTER,BOTTOM,OTHER/NK SPECIES	Trawl_Bottom
59	TRAWL,OTTER,BOTTOM,OTHER/NK SPECIES	Trawl_Bottom
59	TRAWL,OTTER,BOTTOM,OTHER/NK SPECIES	Trawl_Bottom
59	TRAWL,OTTER,BOTTOM,OTHER/NK SPECIES	Trawl_Bottom
60	TROLL LINE, OTHER/NK SPECIES	Trawl_Bottom

65	POWER TROLL OTHER/NK SPECIES	Trawl_Bottom
70	HAUL SEINE, BEACH, COMMON	Seine
70	HAUL SEINE, BEACH, COMMON	Seine
80	FLOATING TRAP	Pot/Trap
90	DIP NET, COMMON	Net
	GILL NET, FIXED OR ANCHORED,SINK, OTHER/NK	
100	SPECIES	Gillnet
120	PURSE SEINE, OTHER/NK SPECIES	Seine
132	DREDGE, SCALLOP,SEA	Dredge_Scallop
132	DREDGE, SCALLOP,SEA	Dredge_Scallop
132	DREDGE, SCALLOP,SEA	Dredge_Scallop
132	DREDGE, SCALLOP,SEA	Dredge_Scallop
132	DREDGE, SCALLOP,SEA	Dredge_Scallop
140	POUND NET, OTHER/NK SPECIES	Net
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
180	POTS + TRAPS,OTHER/NK SPECIES	Pot/Trap
200	POT/TRAP, LOBSTER OFFSH NK	Pot/Trap
210	POT/TRAP, LOBSTER INSH NK	Pot/Trap
210	POT/TRAP, LOBSTER INSH NK	Pot/Trap
250	RAKES, OTHER/NK SPECIES	Handgear
250	RAKES, OTHER/NK SPECIES	Handgear
260	WEIR	Handgear
260	WEIR	Handgear
260	WEIR	Handgear
300	POTS + TRAPS,CRAB OTHER	Pot/Trap
300	POTS + TRAPS,CRAB OTHER	Pot/Trap
301	POTS + TRAPS,BLUE CRAB	Pot/Trap
350	BEAM TRAWL, OTHER/NK SPECIES	Trawl_Bottom
386	DREDGE, CLAM, HYDRAULIC	Dredge_Hydraulic
410	TONGS AND GRABS, OTHER/NK SPECIES	Handgear
420	HOOKS	Handgear
420	HOOKS	Handgear
530	TRAMMEL NET	Net
530	TRAMMEL NET	Net
530	TRAMMEL NET	Net
530	TRAMMEL NET	Net
530	TRAMMEL NET	Net
530	TRAMMEL NET	Net
999	UNKNOWN	Unknown
999	UNKNOWN	Unknown
999	UNKNOWN	Unknown
999	UNKNOWN	Unknown
999	UNKNOWN	Unknown
999	UNKNOWN	Unknown
